Relaxation between the two states

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I. INTRODUCTION

In this note we reveal the typical form of the coupling rate and compare against the phenomenological formula.

II. MODEL

We consider two state $|n\rangle$ and $|m\rangle$, which stands for the the two charge configurations. Addtionally there is a coupling between the electron states and the environment of the form $\hat{V}(x) = \hat{V}_s(x) \otimes \hat{V}_e(x)$, where V_e acts on the environment and V_s acts only on the system. The states are separated in energy by the Hamiltonian $H_0 = E(n,v)\,|n\rangle\langle n| + E(m,v)\,|m\rangle\langle m|$, and the environment has Hamiltonian $H_e = \sum_\alpha E_\alpha\,|\alpha\rangle\langle\alpha|$. We compute now probablity of transition from state $|n\rangle$ to $|m\rangle$, which is given by:

$$p_{mn}(t) = \sum_{\alpha\beta} \left| \langle \alpha, m|\, \hat{U}_I(t)\, |\beta, n\rangle \right|^2 \tag{1}$$

where the environment is in state $\varrho = \sum_\alpha |\alpha\rangle\langle\alpha|\, p_\alpha$, where $|\alpha\rangle$ is some basis, and the unitary operator in interaction picture is given by

$$\hat{U}_I(t) = \hat{T} \exp\left(-i \int_0^t e^{i\hat{H}_0 t'} \hat{V}_e(x) e^{-i\hat{H}_0 t'} dt' \right). \tag{2}$$

We now look at the leading order correction, i.e. $\hat{U}_I(t) = 1 - \frac{i}{\hbar}\int_0^t dt' e^{i(\hat{H}_0+\hat{H}_e)t'} \hat{V}(x) e^{-i(\hat{H}_0+\hat{H}_1)t'}$ and split the interaction $\hat{V}_e(x) = \hat{V}_s(x) \otimes \hat{V}_e(x)$, where V_s acts on the system only, and $V_e(x)$ only on the environment. This

We now define spectral density of the environment as:

$$S(\omega) = \int_{-\infty}^{\infty} \mathrm{d}t e^{i\omega t} C(t) \tag{4}$$

where $C(t) = \mathrm{Tr}_e\{V_e(t)V_e(0)\varrho_e\}$. Which allows to write the double integral as:

$$p_{mn}(t) = \left| \langle m|\, \hat{V}_s(x)\, |n\rangle \right|^2 \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi} S(\omega) \left| \int_0^t \mathrm{d}t_1 e^{-i(\omega - E_{mn})t_1} \right|^2 \tag{5}$$

where $E_{mn} = E_m - E_n$. We now compute the integral, which gives:

$$p_{mn}(t) = t \left| \langle m|\, \hat{V}_s(x)\, |n\rangle \right|^2 \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi} S(\omega) t \mathrm{sinc}^2\left(\frac{\omega - E_{mn}}{2} t \right) \tag{6}$$

Finally we assume t is sufficiently large such that the $t\mathrm{sinc}^2(at/2) \to 2\pi\delta(a)$, which gives:

$$\Gamma_{mn}(t) \equiv p_{mn}/t = \left| \langle m|\, \hat{V}_s(x)\, |n\rangle \right|^2 S(E_{mn}) \tag{7}$$

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substituted gives:

$$\begin{aligned}
p_{mn}(t) = &\int_0^t \mathrm{d}t_1 \mathrm{d}t_2 \sum_{\alpha\beta} p_\alpha \\
&\langle m|\, e^{i\hat{H}_0 t_1} \hat{V}_s(x) e^{-i\hat{H}_0 t_1}\, |n\rangle\, \langle \alpha|\, \hat{V}_e(x, t_1)\, |\beta\rangle \\
&\langle n|\, e^{i\hat{H}_0 t_2} \hat{V}_s^\dagger(x) e^{-i\hat{H}_0 t_2}\, |m\rangle\, \langle \beta|\, \hat{V}_e^\dagger(x, t_2)\, |\alpha\rangle \\
&= \left| \langle m|\, \hat{V}_s(x)\, |n\rangle \right|^2 \int_0^t \mathrm{d}t_1 \mathrm{d}t_2 e^{i(E_m - E_n)(t_1 - t_2)} \\
&\times \sum_\alpha p_\alpha\, \langle \alpha|\, \hat{V}_e(x, t_1) \sum_\beta |\beta\rangle\langle\beta|\, \hat{V}_e(x, t_2)\, |\alpha\rangle \\
&= \left| \langle m|\, \hat{V}_s(x)\, |n\rangle \right|^2 \int_0^t \mathrm{d}t_1 \mathrm{d}t_2 e^{i(E_m - E_n)(t_1 - t_2)}
\end{aligned} \tag{3}$$

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